

ScenePulse

Master Phase-Wise Build Specification

Replit AI Agent + Supabase + n8n Edition

Version: Hackathon MVP + Startup Foundation

PROJECT OVERVIEW

ScenePulse is an AI-powered local music intelligence platform that helps fans discover artists, musicians find collaborators, venues discover talent, and labels identify emerging artists.

The platform combines:

- Interactive Music Map
- AI Music Discovery
- Audio Intelligence
- Lyrics Intelligence
- Live Event Intelligence
- Artist Analytics
- Collaboration Matching
- AI Scene Radio
- Automated Ecosystem Workflows

The application should feel like:

Spotify × Google Maps × LinkedIn × Arc Browser

The design should be modern, premium, minimal, and highly interactive.

TECH STACK

Frontend

- Next.js 15 App Router
- TypeScript
- TailwindCSS
- ShadCN UI
- Framer Motion
- Mapbox GL

Backend

- Supabase
- PostgreSQL
- PostGIS
- pgvector
- Supabase Auth
- Supabase Storage
- Supabase Realtime

Automation

- n8n

AI

- OpenAI
- ElevenLabs
- Cyanite

Music APIs

- Musixmatch
- Songstats
- JamBase
- LALAL.AI

Deployment

- Vercel
- Supabase
- n8n Cloud

PHASE 1

FOUNDATION & INFRASTRUCTURE

Goal

Establish architecture and core project setup.

Expected Outcome

Working application shell with authentication and map.

Frontend Tasks

Create:

/login

/signup

/dashboard

/map

/profile

/settings

Build:

Navigation

Command Bar

Sidebar

Map Container

Search Component

Global Layout

Theme System

Supabase Setup

Enable:

Auth

Storage

Realtime

PostGIS

pgvector

Database Schema

profiles

id

email

role

avatar_url

created_at

artists

id

profile_id

artist_name

bio

spotify_url

latitude

longitude

verified

created_at

venues

id

name

address

latitude

longitude

capacity

created_at

events

id

venue_id

artist_id

event_name

event_date

status

created_at

Authentication

Roles:

fan

artist

venue

label

admin

n8n Workflows

Workflow 1

User Signup

Trigger:

Supabase User Created

Actions:

Create Profile

Assign Role

Send Welcome Email

Workflow 2

Artist Creation

Trigger:

Artist Record Created

Actions:

Initialize Artist Profile

Queue Enrichment Process

Deliverables

Authentication

Map Page

Database

User Roles

n8n Connected

PHASE 2

ARTIST DISCOVERY ENGINE

Goal

Create intelligent artist profiles.

Expected Outcome

Users can discover artists using metadata.

Artist Profile System

Fields

Artist Name

Bio

Genres

Mood Tags

Social Links

Location

Verification Status

Cyanite Integration

Analyze:

Genre

Energy

Mood

BPM

Danceability

Acousticness

Valence

Musixmatch Integration

Analyze:

Lyrics

Themes

Sentiment

Keywords

Language

New Tables

artist_audio_analysis

artist_lyric_analysis

artist_tags

artist_embeddings

n8n Workflow

Workflow 3

Artist Enrichment Engine

Trigger

Spotify URL Added

Process

Fetch Spotify Metadata

↓

Cyanite Analysis

↓

Musixmatch Analysis

↓

Generate AI Summary

↓

Store Results

Workflow 4

Daily Refresh

Trigger

Daily

Process

Refresh Metadata

Update Analytics

Deliverables

Rich Artist Profiles

AI Tags

Mood Search

Genre Search

PHASE 3

LIVE MUSIC INTELLIGENCE

Goal

Transform map into a living ecosystem.

Expected Outcome

Real-time events and artist growth data.

JamBase Integration

Pull:

Events

Venues

Schedules

Tour Data

Songstats Integration

Pull:

Spotify Listeners

Playlist Reach

Growth Rate

Audience Trends

New Tables

artist_metrics

venue_metrics

event_metrics

trend_data

Map Features

Event Pins

Venue Pins

Artist Pins

Cluster Groups

Dynamic Filters

n8n Workflow

Workflow 5

Event Synchronization

Every 30 Minutes

JamBase



Normalize Data



Insert Events



Update Map

Workflow 6

Artist Metrics Sync

Every Hour

Songstats



Fetch Metrics



Calculate Growth



Store Results

Workflow 7

Trending Artists

Every Hour

Calculate Momentum Score

Update Rankings

Deliverables

Live Event Discovery

Artist Analytics

Venue Discovery

Trend Tracking

PHASE 4

COLLABORATION NETWORK

Goal

Create network effects.

Expected Outcome

Artists can find collaborators.

Collaboration Features

Create Request

Accept Request

Decline Request

Direct Message

Recommendation Engine

Collaboration Types

Producer

Vocalist

Guitarist

Bassist

Drummer

DJ

Songwriter

Mix Engineer

Mastering Engineer

New Tables

collaborations

collaboration_requests

artist_connections

artist_matches

Matching Engine

Inputs

Genre

Mood

Energy

Location

Experience

Popularity

pgvector Search

Store Embeddings

Artist Summary

Lyrics Summary

Mood Summary

Audio Summary

n8n Workflow

Workflow 8

AI Match Engine

Trigger

Profile Updated

Process

Generate Embeddings

Find Similar Artists

Rank Matches

Store Results

Workflow 9

Venue Talent Matching

Trigger

Venue Creates Event

Process

Find Nearby Artists

Score Candidates

Recommend Top 20

Deliverables

Collaboration Requests

Match Recommendations

Venue Talent Discovery

PHASE 5

AI DISCOVERY ENGINE

Goal

Natural language music search.

Expected Outcome

Users search using conversational language.

AI Search Examples

Find emotional indie artists

Find artists like Coldplay

Find heartbreak songs

Show jazz musicians near me

Find rising electronic producers

OpenAI Layer

Generate

Embeddings

Recommendations

Summaries

Search Responses

Search Flow

User Query

↓

Embedding Generation

↓

Vector Search

↓

Artist Retrieval

↓

Result Ranking

↓

Map Display

n8n Workflow

Workflow 10

Search Enrichment

Generate Search Metadata

Optimize Recommendations

Deliverables

Semantic Search

Vector Search

AI Recommendations

PHASE 6

SCENE RADIO

Goal

Create memorable demo experience.

Expected Outcome

Interactive AI radio station.

ElevenLabs Integration

Generate

Voice DJ

Artist Introductions

Event Announcements

Weekly Reports

Radio Features

Play Button

Live Feed

Scene Updates

Trending Artists

Event Highlights

Database Tables

radio_broadcasts

radio_scripts

radio_analytics

n8n Workflow

Workflow 11

Scene Radio Generator

Hourly

Fetch Events

↓

Fetch Trends

↓

Generate Script

↓

ElevenLabs

↓

Store Audio

↓

Publish Feed

Deliverables

AI DJ

Audio Feed

Scene Announcements

PHASE 7

HEATMAP & INTELLIGENCE LAYER

Goal

Visualize local music ecosystems.

Expected Outcome

A&R style intelligence.

Heatmap Types

Artist Density

Genre Density

Venue Density

Growth Density

Event Density

New Tables

heatmaps

heatmap_snapshots

regional_trends

breakout_predictions

n8n Workflow

Workflow 12

Heatmap Generator

Hourly

Analyze Data

Update Heatmaps

Workflow 13

Breakout Artist Detector

Daily

Growth Analysis

Trend Analysis

Prediction Score

Deliverables

Music Intelligence Layer

Predictive Analytics

Trend Detection

UI REQUIREMENTS

Design Inspiration

Linear

Spotify

Arc Browser

Apple Maps

Vercel

Notion

Design System

Glassmorphism

Large Typography

Soft Shadows

Motion-Based UX

Microinteractions

Gradient Accents

Minimal Navigation

Dark Mode First

PAGE STRUCTURE

Landing Page

Map Page

Artist Profile

Venue Profile

Event Profile

Collaboration Hub

AI Search

Scene Radio

User Dashboard

Settings

Admin Dashboard

HACKATHON DEMO FLOW

1 Open Landing Page

2 Login

3 Open Interactive Map

4 Search:
"Emotional Indie Artists"

5 Display AI Search Results

6 Open Artist Profile

7 Show

Mood Analysis

Lyrics Themes

Growth Metrics

Spotify Data

8 Show Nearby Events

9 Open Scene Radio

10 Play AI DJ

11 Create Collaboration Request

12 Show Recommended Artists

13 Show Heatmap

14 Show Breakout Artist Prediction

15 End Presentation

SUCCESS CRITERIA

Users can:

Discover Artists

Discover Events

Analyze Music

Find Collaborators

Listen to AI Radio

Receive Recommendations

Explore Heatmaps

Track Growth

Everything updates automatically through n8n workflows and Supabase realtime subscriptions.

ScenePulse should feel like a living music operating system rather than a traditional music directory.